DCRP & PDCP

Enterprise Integration Architecture for SAP BTP

Eliminating Proxy Sprawl and Package Sprawl at Scale

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Executive Summary

Solving Package Sprawl & Governance Chaos in SAP BTP

SAP BTP

The Problem

"Package Sprawl" Crisis

Uncontrolled proliferation of 100+ unstandardized packages in SAP CPI leading to operational chaos.

- Operational Debt & "Silo" mindset
- Credential Hell (1 user per package)
- Tenant Capacity Crises
- Complex Maintenance Overhead

The Solution

DCRP + PDCP Architecture

Consolidates integrations into governed packages per business domain with metadata-driven routing.

DCRP: Domain-Centric Routing Pattern (Gateway Layer)

PDCP: Package Domain-Centric Pattern (Backend Layer)

Tiered Users: Segregated by criticality

Validated Impact

API Proxies	40 → 4	↓90%
CPI Packages	40 → 4	↓90%
Technical Users	39 → 12	↓69%
Deployment Time	273m → 14.5m	18.8×

**Validated data from live enterprise implementation (Feb 2026)*

Source: Viana, R. L. H. (2026). Gateway Domain-Centric Routing. Zenodo. DOI: 10.5281/zenodo.18582469

The Package Sprawl Crisis

Visualizing the "Technical Silo" Graveyard in SAP CPI

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Inconsistent Naming

Packages named after projects, temporary needs, or vendors. Impossible to search or govern. "Where is the Sales logic?"

Redundant Governance

Multiple packages handling the same domain (e.g., Salesforce_V1, Sales_New, Orders_Old). Fragmented logic across 100+ locations.

Credential Chaos

One technical user per package leads to credential hell. 39 packages = 39 users to manage, rotate, and secure.

Typical "Sprawl" Tenant Structure

```
CPI Tenant Root
  Package Salesforce.Integration
  Salesforce_Orders_NEW_FINAL
  VTEX.Flow
  MercadoLivre_Shipments_API
  D365.Finance.Payments.v2.0
  D365-Invoices
  ServiceNow_Incidents_Sync
  Logistics_3PL_Freight_v3
  SAP__MasterData_Customer_v2
  Temp_Testing_DoNotDelete_QAS
  Dev.Customer.Old.Logic
  [+147 more packages... (Legacy debt)]
```

100+
Unmanaged Packages

Introduction to DCRP (Gateway)

Domain-Centric Routing Pattern: The Intelligent Facade

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The Core Concept

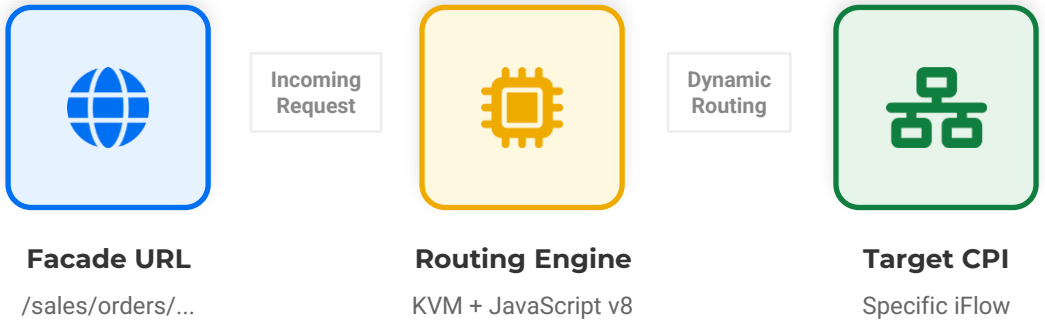
Instead of creating one API Proxy per integration (1:1), DCRP implements a **single intelligent gateway per business domain**. It uses metadata to dynamically route requests to the correct backend endpoint.

Ends "Proxy Sprawl" Forever

Zero-Code Onboarding (Config Only)

Centralized Security Policies

LOGICAL ARCHITECTURE



1. Extract domain & vendor
2. Query Key-Value Map (KVM)
3. Decode target URL

```
route = kvm.idintegration = dcrporderscsalesforceid01:http,dcrpordersusalesforceemeaid02:http
```

Introduction to PDCP (Backend)

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Package Domain-Centric Pattern: Organized Backend Governance

The Core Principle

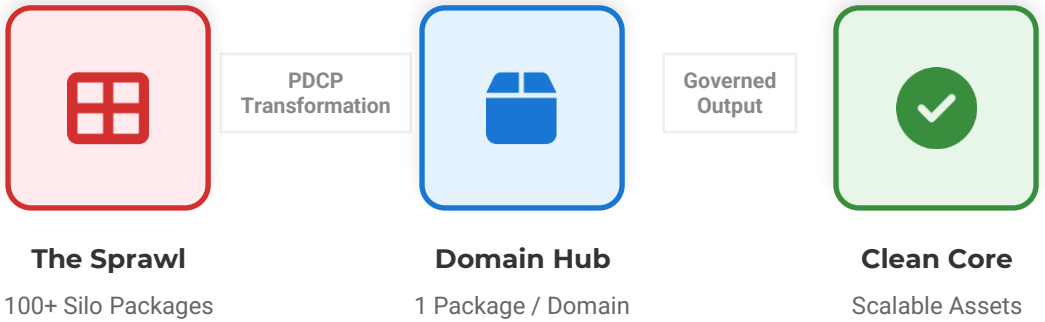
Instead of creating a package for every application (e.g., "Salesforce"), PDCP consolidates integration assets into **one single package per business domain** (e.g., "Sales"). It shifts focus from *"Where is it from?"* to *"What is the business doing?"*.

Consolidation: 39 Packages → 4 Domains

Standardization: iFlow DNA for indexing

Governance: Domain Architect Gatekeeper

TRANSFORMATION LOGIC



Format: [company].[domain].[subprocess].integrations

`nx.sales.o2c.integrations`

(Contains all orders, customers, invoice flows)

The Complete Architecture: DCRP + PDCP

End-to-End Flow: From Consumer to Governed Backend Domain

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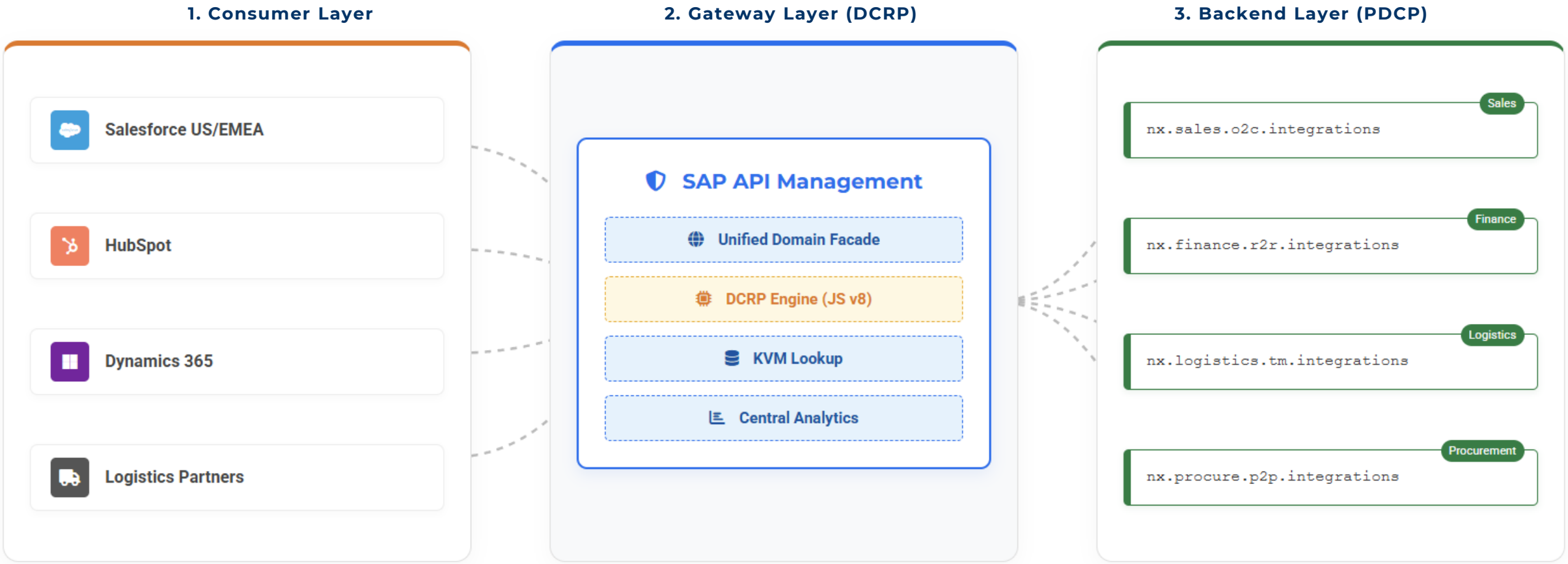

13 → 1	39 → 4	39 → 12	HTTPS / SOAP / ODATA	18.8×
API Proxies	CPI Packages	Tech Users	Supported Protocols	Faster Deploy

The Three Pillars of PDCP

Foundational Principles for a Scalable Integration Backend

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1

Consolidation

Mandatory grouping of integration assets by Business Domain rather than source system or project.

- Group by **Business Area** (Sales, Finance, Logistics)
- Prohibits "Project-Specific" packages
- Eliminates redundant "Vendor-Specific" silos

2

Standardization

Implementation of the **iFlow DNA** naming convention for instant discovery and automated categorization.

- Unique iFlow Identifiers (id01, id02...)
- Standardized Action Codes (c, u, s, n)
- Enables instant ALM indexing & search

3

Governance

Establishing the Domain Architect as a gatekeeper to ensure architectural integrity over time.

- Prevents creation of new silos
- Enforces naming compliance before deploy
- Manages lifecycle of tiered users

Consolidation Metrics & ROI

Quantifying the Impact of PDCP Architecture

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Metric	Before	After	Improvement
	39	**4**	↓ 90%
	~390	**13**	↓ 97%
	~39	**12**	↓ 69%
	273 min	**14.5 min**	18.8× Faster
	10 min	**5 sec**	120× Faster

224.5h

Annual Time Saved

€ 198.5k

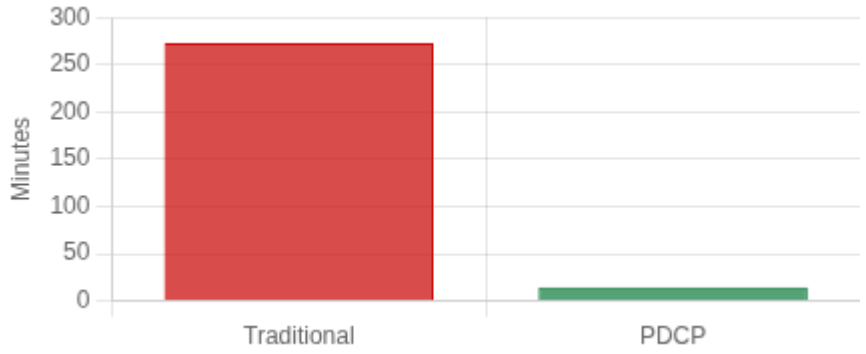
5-Year Value Projection

Deployment Efficiency Formula

$$T = (N_{pkg} \times 2m) + (N_{iFlows} \times 0.5m)$$

Before: (39 × 2) + (390 × 0.5) = **273 min**
After: (4 × 2) + (13 × 0.5) = **14.5 min**

Deployment Time Reduction



iFlow DNA: Naming Convention

Standardized 7-Segment Format for Surgical Indexing

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Standard Format Structure

```
id[seq] . [subdomain] . [sender] . [entity] . [receiver] . [action] . [direction] . [sync/async]
```

Component Breakdown

Segment	Meaning	Examples
id[seq]	Unique Index	id01, id02, id03 (Resets per subdomain)
subdomain	Business Process	o2c (Order-to-Cash), r2r (Record-to-Report)
sender	Source System	salesforce, shopify, s4hana, magento
entity	Business Object	order, customer, invoice, payment
receiver	Target System	s4hana, salesforce, quickbooks, fedex
action	Function Code	c (create), u (update), s (sync) - see table
direction	Data Flow	in (Inbound), out (Outbound)

Standard Action Codes

c	Create	r	Read
u	Update	d	Delete
s	Sync	n	Notify
t	Transform	a	Approve

Practical Example

```
id01.o2c.salesforce.order.s4hana.c.in.sync
```

"First interface in Order-to-Cash, creating an Order from Salesforce to S/4HANA (Inbound, Synchronous)"

Technical User Strategy

Tiered Credentials for Risk Containment & Audit Precision

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The Risk: Single Point of Failure

Using one "superuser" for an entire package means a password lockout or attack kills **all interfaces** simultaneously (Core + Support).

The PDCP Solution

- **Blast Radius Control:** Isolate critical flows from non-critical failures.
- **Audit Precision:** "401 Unauthorized" instantly reveals *which* tier failed.
- **Vendor Isolation:** Protect internal systems from 3rd-party erratic behavior.

Tiered User Architecture (Sales Domain Example)

Tier 1: Transactional Core

`cr.so.o2c`

Mission-critical flows (Order Creation). Highest monitoring priority. Isolated from support noise.

Tier 2: Support / Master Data

`cr.so.md`

Secondary flows (Customer Sync, Price Updates). Failures here don't stop revenue.

Tier 3: Third-Party / Vendor

`cr.so.vendor`

External integrations prone to lockouts. Strictly isolated from internal core credentials.

Vendor Lockout	→	Core O2C	→	Master Data
FAILED		RUNNING		RUNNING

Real Implementation Example

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Domain Package: Sales Order-to-Cash (13 Consolidated iFlows)

TARGET PACKAGE: `nx.sales.o2c.integrations`

25 min Deploy Time

ID	Sender	Entity	Receiver	Act	Dir	Proc	Proto	SAP CPI Endpoint Pattern
id01	salesforce	orders	s4hana	c	in	sync	REST	/http/dcrp/orders/c/id01
id02	salesforce	orders	s4hana	u	in	sync	REST	/http/dcrp/orders/u/id02
id03	shopify	customers	s4hana	s	in	sync	REST	/http/dcrp/customers/s/id03
id04	stripe	payments	s4hana	n	in	sync	REST	/http/dcrp/payments/n/id04
id05	microsoft	orders	s4hana	c	in	sync	SOAP	/cxf/dcrp/orders/c/id05
id06	fedex	deliveries	s4hana	t	in	sync	REST	/http/dcrp/deliveries/t/id06
id07	s4hana	customers	externalcrm	s	out	sync	SOAP	/cxf/dcrp/customers/s/id07
id08	s4hana	payments	externalpay	n	out	sync	SOAP	/cxf/dcrp/payments/n/id08
id09	quickbooks	invoices	s4hana	c	in	sync	SOAP	/cxf/dcrp/invoices/c/id09
id10	s4hana	invoices	externalbill	c	out	sync	REST	/http/dcrp/invoices/c/id10
id11	s4hana	deliveries	externallog	t	out	sync	SOAP	/cxf/dcrp/deliveries/t/id11
id12	shopify	returns	s4hana	c	in	sync	REST	/http/dcrp/returns/c/id12

Endpoint patterns (/http/dcrp/... vs /cxf/dcrp/...) are critical for DCRP JavaScript routing logic

URL Governance Patterns

Defining Agnostic Endpoints for Maximum Traceability

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Option 1: Full Context

Recommended

URL Format Structure

```
/[entity]/[action]/[id]
```

Implementation Examples

- `/orders/c/id01`
- `/customers/s/id03`
- `/payments/n/id04`
- `/deliveries/t/id06`

Key Benefits

- Full context in URL (Entity + Action)
- Direct mapping to iFlow name structure
- Easiest troubleshooting via URL ID

Option 2: Simplified

URL Format Structure

```
/[entity]/[id]
```

Implementation Examples

- `/orders/id01`
- `/customers/id03`
- `/payments/id04`
- `/deliveries/id06`

Key Benefits

- Cleaner, shorter URLs
- Logical entity-based grouping
- ID ensures uniqueness without action code

Option 3: Domain Prefix

URL Format Structure

```
/sales/[entity]/[action]/[id]
```

Implementation Examples

- `/sales/orders/c/id01`
- `/sales/customers/s/id03`
- `/sales/payments/n/id04`
- `/sales/deliveries/t/id06`

Key Benefits

- Explicit domain-level organization
- Clear business context at root level
- Scalable across multiple business units

DCRP + PDCP Routing Flow

5-Step Execution from Facade URL to Backend Process

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1 Facade URL

Client calls agnostic facade with vendor identifier:

```
POST
/sales/orders/creation/salesforce
```

Explicit vendor ensures correct routing

2 Conversion

Gateway converts facade URL to routing key:

```
/sales/orders/creation/salesforce
↓
dcrporderscid01
```

Removing slashes creates unique KVM key

3 KVM Lookup

Query Key-Value Map for metadata:

Key:

`Dcrporderscid01:http`

Value:

`:http`

Retrieves adapter type (:http or :cxf)

4 JS Engine

v8.0 Engine constructs target URL:

```
var adapter = "http";
var path = "/dcrp/orders/c/id01";
target = host+"/"+ adapter +path;
```

`/http/dcrp/orders/c/id01`

Generates the CPI Endpoint

5 CPI Execution

CPI receives request and executes mapped iFlow:

`id01.o2c.salesforce.order.c.in.sync`

Exact match execution

This automated flow enables zero-code vendor onboarding by simply adding KVM entries

KVM Configuration Structure

Metadata-Driven Routing Logic for DCRP Gateway

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KVM Field Structure (Per Domain)

```
cpipackage-nx.sales.o2c.integrations
  <Put enabled="true">
    <Key>packagename</Key>
    <Value>nx.sales.o2c.integrations</Value>
  </Put>
  <Put enabled="true">
    <Key>saprocess</Key>
    <Value>SAP SD & O2C - Sales Process</Value>
  </Put>
  <Put enabled="true">
    <Key>idinterface</Key>
    <Value>dcrporderscid01:http,dcrporderscid05:cxf...</Value>
  </Put>
</KeyValueMapOperations>
```

packagename	Matches the PDCP package name in CPI backend
saprocess	Business process description for documentation/logging
idinterface	**Core Routing Map:** Comma-separated list of [key]:[adapter] pairs

Adapter Type Logic

Defines the target endpoint prefix for the JS engine:

:http	:cxf
`/http/dcrp/...`	`/cxf/dcrp/...`
REST / OData	SOAP

Default: If omitted, defaults to :http

Map Identifier Naming

Pattern must mirror package name:

`cpipackage-[package_name]`

Examples:

- ✓ cpipackage-nx.sales.o2c.integrations
- ✓ cpipackage-nx.finance.r2r.integrations
- ✓ cpipackage-nx.logistics.tm.integrations

Multi-Vendor Disambiguation

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Handling Complexity: Regions, Environments & Platforms

Salesforce Multi-Region

Process: Order Creation

Vendor Instance	Target iFlow
salesforceus	id01...c.in.sync
salesforceeu	id02...c.in.sync
salesforceapac	id03...c.in.sync

KVM Structure Example

```
dcrporderscid01:http,
dcrporderscid02:http,
dcrporderscid03:http
```

S/4HANA Environments

Process: Invoice Outbound

Environment	Target iFlow
s4prod	id01...c.out.sync
s4qa	id10...c.out.sync

KVM Structure Example

```
dcrpinvoicescid01:cxf,
dcrpinvoicescid10:cxf
```

E-Commerce Platforms

Process: Order Inbound

Platform	Target iFlow
shopify	id04...c.in.sync
magento	id08...c.in.sync
vtex	id16...c.in.sync

KVM Structure Example

```
dcrporderscid04:http,
dcrporderscid16:cxf
```

- Granular Vendor Metrics
- Isolated Error Handling
- Vendor-Specific Security Policies
- Zero-Code Onboarding

DCRP enables "Vendor Awareness" without complex hardcoded logic in APIM

API Proxy Configuration

Single Gateway Facade for Domain Consolidation

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Proxy Definition

`cpi-dcrp-proxy-domain-sales-o2c` `/sales` `HTTP/S`

Route Rules (13 Entities)

All routes point to same Target Endpoint

Route Name	Condition (Path Suffix)	Target Flow
sales-o2c-orders	MatchesPath "/orders/**"	*default*
sales-o2c-customers	MatchesPath "/customers/**"	*default*
sales-o2c-payments	MatchesPath "/payments/**"	*default*
sales-o2c-invoices	MatchesPath "/invoices/**"	*default*
sales-o2c-deliveries	MatchesPath "/deliveries/**"	*default*
sales-o2c-returns	MatchesPath "/returns/**"	*default*
sales-o2c-quotes	MatchesPath "/quotes/**"	*default*
+ 6 more rules (pricing, credits, collections, contracts, disputes, revenue)		

Overhead Impact Analysis

API Proxies	13	1
Security Policies	13	1
Dashboards	13	1
Maintenance Objects	High	Low

92% Reduction in Overhead

Testing & Validation Results

Verified Performance & Reliability Metrics

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Test Execution (Sales O2C)

Environment	Collection	Auth Method
SAP BTP (Cloud Foundry)	DCRP-PDCP-Sales-O2C	OAuth 2.0 Client Creds

Test Case Name	Facade Endpoint	Status
Order Creation - Salesforce	POST /orders/creation/salesforce	✓ PASS 200
Order Update - Salesforce EMEA	PUT /orders/update/salesforceemea	✓ PASS 200
Customer Sync - Shopify	POST /customers/sync/shopify	✓ PASS 200
Payment Notification - Stripe	POST /payments/notification/stripe	✓ PASS 200
Delivery Tracking - FedEx	POST /deliveries/tracking/fedex	✓ PASS 200
Invoice Creation - QuickBooks	POST /invoices/creation/quickbooks	✓ PASS 200
Return Creation - Shopify	POST /returns/creation/shopify	✓ PASS 200
Invoice Creation - S/4HANA	POST /invoices/creation/s4hana	⚠ CONFLICT 404
+ 5 additional passing tests (id05, id07, id08, id11, id13)		

Note: Conflict on id09/id10 due to identical URL patterns (Resolved in Part III with JS logic)

Performance KPIs

33k+
Messages Tested

68ms
Avg. Latency

Overall Success Rate

11 Passed / 2 Conflicts
(100% Success in Dedicated Tests)

Scalability Projection

Growth Impact Analysis: +100 Integrations Scenario

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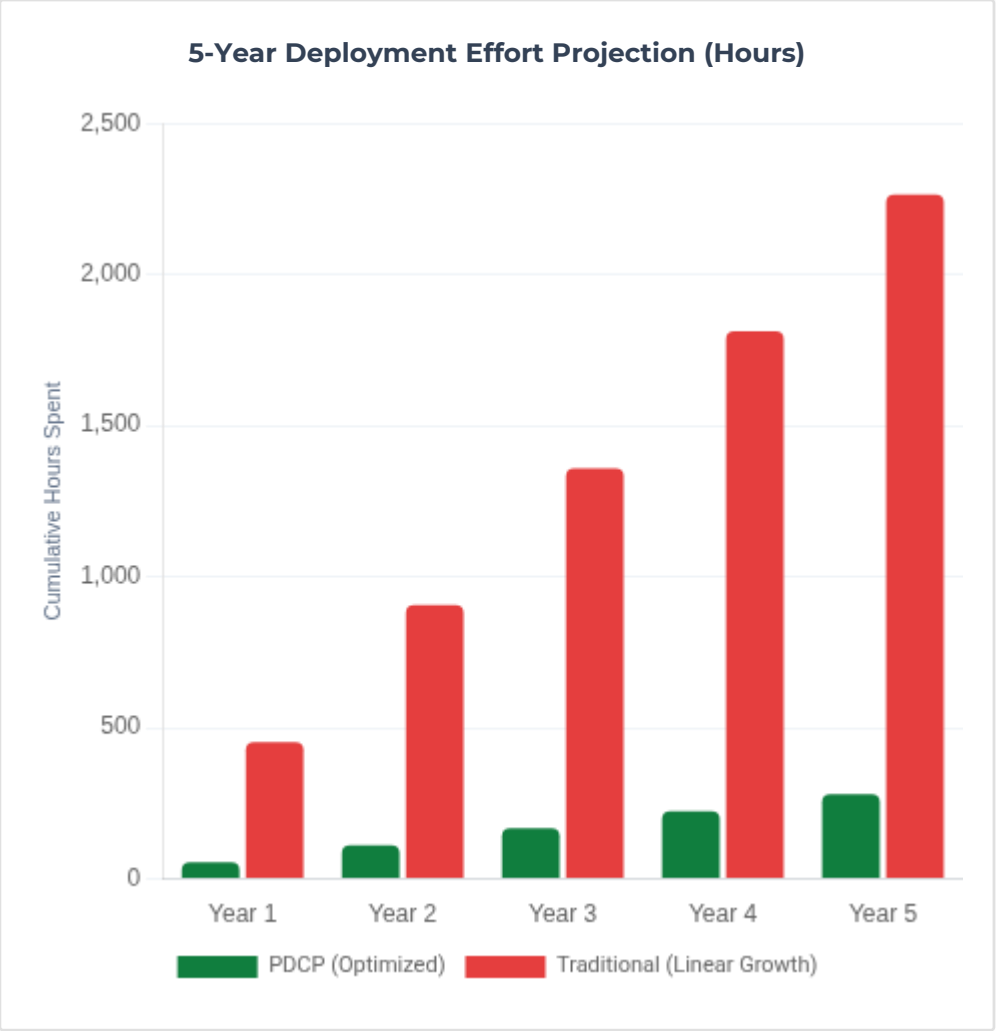
Growth Scenario (+100 Integrations)

Metric	Traditional	PDCP	Difference
Packages	39 → 139	4 → 4	+0 Growth
Tech Users	39 → 139	12 → 24	Tiered Strategy
Deploy Time	273 → 523 min	14.5 → 64.5 min	8× Faster
Annual Hours*	453h	56h	397h Saved

Time Impact Calculation

Annual Savings: 397 hours/year saved in deployment labor alone.
Business Value: At €100/hour engineering cost = **€39,700/year** savings (excluding incident costs).

PDCP scales sub-linearly: 5x volume = only 1.2x overhead



*Based on 52 deployments/year frequency. Traditional model scales linearly; PDCP stabilizes overhead.

Implementation Roadmap

Phased Execution Plan: From Consolidation to Automation

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Phase 1: PDCP Foundation

Weeks 1-4

1. **Domain Mapping**
   Analyze landscape and define the 4-6 core business domains (Sales, Finance, Logistics, etc.).
2. **Package Consolidation**
   Create 1 package per domain (e.g., nx.sales.o2c.integrations) and consolidate existing iFlows.
3. **Apply iFlow DNA**
   Rename all assets using the 7-segment standard: idXX.[sub].[sender].[entity]...
4. **Tiered Security**
   Implement 3-tier user strategy (Core, Master Data, Vendor) for risk isolation.

Milestone: Clean Backend Core

Phase 2: DCRP Gateway

Weeks 5-8

5. **Proxy Creation**
   Deploy single API Proxy per domain in SAP APIM (e.g., cpi-dcrp-proxy-domain-sales-o2c).
6. **KVM Configuration**
   Set up Key-Value Maps linking compact routing keys to iFlow endpoints.
7. **JavaScript v8 Engine**
   Deploy the dynamic routing script to handle URL decoding and vendor logic.
8. **Gradual Migration**
   Switch consumers to new Facade URLs progressively (vendor by vendor).

Milestone: Intelligent Routing

Edge Case Handling

- **Hotfixes:** Direct iFlow deploy supported via granular naming
- **PCI/Security:** Policy enforcement per route path
- **Quotas:** Rate limiting applied at proxy route level

Conclusions & Next Steps

From Technical Debt to Strategic Asset

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Core Principles

- **Order by Design:** Replaced 39+ fragmented packages with 4 governed domains, reducing governance objects by 90%.
- **DCRP + PDCP Synergy:** Gateway abstraction layer combined with standardized backend execution ensures scalability.
- **Metadata-Driven:** Zero-code vendor onboarding via KVM configuration instead of proxy development.

Validated Gains

↓90%	↓69%	18.8×
Packages	Tech Users	Deploy Velocity

Recommended Actions

01 Validate Domains & Naming Taxonomy

02 Prototype 'Sales O2C' (nx.sales.o2c)

03 Measure KPIs & Scale to Finance/Logistics

Academic Reference

Viana, R. L. H. (2026). Gateway Domain-Centric Routing: A Vendor-Agnostic Metadata-Driven Architecture for Enterprise API Governance.

"Order is your greatest tactical advantage. A clean backend allows an enterprise to move at the speed of business." – Viana | The Commander